Stable Menus of Public Goods

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Motivation

Decision makers often select "public goods" to provide to unit-demand agents:

- Which electives should an online school offer?
- Where should a vending machine company locate identical machines?
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- View as matching problem: agents match to favorite available good.
- Each good needs minimum usage to justify existence.
  $\hookrightarrow$ goods' preferences have complementarities
- No capacity constraints (unlike much of the assignment literature).

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This talk:

- How to define stability for a matching in this setting?
- Existence of stable outcomes? Strategic considerations?

Related Work

Matching

Public projects

Committee selection

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- Existence (Gale + Shapley '62)
- No complementarities (Hatfield and Kojima '08)
- Strategyproofness (Dubins + Freedman '81, Roth '82)

This work: no capacity constraints, yes complementarities

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This work: no money

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Committee selection

- Aziz et al. '14, Jiang + Mungala + Wang '20

This work: no budget

Model

- n **agents**, denoted $N = \{1, \ldots, n\}$.
- g **public goods**, denoted $G = \{1, \ldots, g\}$.
- Each agent $i \in \{1, \ldots, n\}$ has **complete preferences** $\succ_i$ over G.
- A **menu** $M \subseteq G$ induces a **matching**: agent i uses their favorite good in M

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A menu $M \subseteq G$ of public goods is t-**stable** if:

- t-**feasibility**: each provided public good $\gamma \in M$ is used by $\geq t$ agents.
- t-**uncontestability**: there do not exist t "unhappy" agents, and an unprovided public good $\gamma \in G \setminus M$, such that each of these agents prefers γ over all provided public goods in M.

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- **Menu selection problem** = (agents, public goods, preferences).

Example

$$\left.\begin{array}{l}\bullet\ \ t\text{-}\textbf{feasibility}\text{: each provided good } \gamma \in M \text{ used by } \geq t \text{ agents.} \\ \bullet\ \ t\text{-}\textbf{uncontestability}\text{: } \nexists\ t \text{ “unhappy” agents who prefer } \gamma \in G \setminus M\end{array}\right\}\ t\text{-}\textbf{stable}$$

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Example

$t = 4$

$n = 9$ agents

$g = 3$ goods

Agents

$\mathbf{3} \times 1 \succ 2 \succ 3$

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No t-stable menu exists!

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$\Big\}$ (t, u)-**stable**

Example

$t = 4$, $u = 7$

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When u/t sufficiently large, (t, u)-stable menus exist.

Overview

We are interested in questions of:

Existence

$\hookrightarrow$ For which t, u do (t, u)-stable menus exist for all menu selection problems?

Strategyproofness

$\hookrightarrow$ When existence guaranteed, for which g, t, u is there a SP mechanism?

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  - Tight characterization for $g \leq 6$

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- $g = 2$: simple anonymous SP mechanism
- $g = 3, 4, 5, 6$: impossibility result (no anonymous SP mechanism)

Existence

Simple bounds

Lower bound

Upper bound

Simple bounds

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Theorem. Let $g \geq 3$ and $u \leq 2t - 2$. Then there exists a menu selection problem with no (t, u)-stable menu.

Upper bound

Simple bounds

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Proof sketch. One can check these agents have no (t, u)-stable menu:

- $t - 1 \times 1 \succ 2 \succ 3$
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Upper bound

Proposition. Let $g \geq 2$ and $u > g(t - 1)$. Then for all menu selection problems, there exists a (t, u)-stable menu.

Simple bounds

Lower bound

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Upper bound

Proposition. Let $g \geq 2$ and $u > g(t - 1)$. Then for all menu selection problems, there exists a (t, u)-stable menu.

Proof sketch. Let $M := \{\gamma \in G : \exists t \text{ agents with favorite good } \gamma\}$. One can check M is (t, u)-stable.

Tight characterization for $g \leq 6$

To guarantee existence of stable menus, simple bounds say:

$$\left.\begin{array}{l} \bullet \text{ Necessary: } u \geq 2t - 1. \\ \bullet \text{ Sufficient: } u \geq g(t-1) + 1. \end{array}\right\} \text{gap of factor of} \sim g$$

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This simple lower bound is tight for $g = 3, 4, 5, 6$:

Theorem. Let $g \in \{3, 4, 5, 6\}$ and $u \geq 2t - 1$. Then every menu selection problem has a (t, u)-stable menu.

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(We'll talk about $g \geq 7$ later...)

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Proof sketch.

- $g = 3, 4$: analyze **greedy algorithm**. Analyzing cycle reveals stable menu.
- $g = 5, 6$: solve **computationally**. Using structural insights, reduce to polyhedra covering problem $\hookrightarrow$ 1 week on Harvard cluster using SMT solver.

Beyond $g \geq 7$

Theorem. Let $g \geq 7$ and $u \leq 23\lfloor \frac{t-1}{11} \rfloor$. Then there exists a menu selection problem with no (t, u)-stable menu. (cf. $u \geq 2t - 1 \Leftrightarrow$ existence when $g \leq 6$)

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Set $x := \lfloor \frac{t-1}{11} \rfloor$. Then the following $70x$ agents have no (t, u)-stable menu:

#	Preference	#	Preference	#	Preference	#	Preference
$5x$	$1 \succ 2 \succ 3$	$3x$	$1 \succ 2 \succ 4 \succ 5$	x	$1 \succ 4 \succ 2 \succ 5$	x	$1 \succ 6 \succ 4 \succ 2$
$5x$	$2 \succ 3 \succ 4$	$3x$	$2 \succ 3 \succ 5 \succ 6$	x	$2 \succ 5 \succ 3 \succ 6$	x	$2 \succ 7 \succ 5 \succ 3$
$5x$	$3 \succ 4 \succ 5$	$3x$	$3 \succ 4 \succ 6 \succ 7$	x	$3 \succ 6 \succ 4 \succ 7$	x	$3 \succ 1 \succ 6 \succ 4$
$5x$	$4 \succ 5 \succ 6$	$3x$	$4 \succ 5 \succ 7 \succ 1$	x	$4 \succ 7 \succ 5 \succ 1$	x	$4 \succ 2 \succ 7 \succ 5$
$5x$	$5 \succ 6 \succ 7$	$3x$	$5 \succ 6 \succ 1 \succ 2$	x	$5 \succ 1 \succ 6 \succ 2$	x	$5 \succ 3 \succ 1 \succ 6$
$5x$	$6 \succ 7 \succ 1$	$3x$	$6 \succ 7 \succ 2 \succ 3$	x	$6 \succ 2 \succ 7 \succ 3$	x	$6 \succ 4 \succ 2 \succ 7$
$5x$	$7 \succ 1 \succ 2$	$3x$	$7 \succ 1 \succ 3 \succ 4$	x	$7 \succ 3 \succ 1 \succ 4$	x	$7 \succ 5 \succ 3 \succ 1$

- Simplified and cleaned from counterexample found by SMT solver.

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$5x$	$3 \succ 4 \succ 5$	$3x$	$3 \succ 4 \succ 6 \succ 7$	x	$3 \succ 6 \succ 4 \succ 7$	x	$3 \succ 1 \succ 6 \succ 4$
$5x$	$4 \succ 5 \succ 6$	$3x$	$4 \succ 5 \succ 7 \succ 1$	x	$4 \succ 7 \succ 5 \succ 1$	x	$4 \succ 2 \succ 7 \succ 5$
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- Simplified and cleaned from counterexample found by SMT solver.
- Also have somewhat improved upper bound: $u \geq (g-2)(t-1) + 1$.

Beyond $g \geq 7$

Theorem. Let $g \geq 7$ and $u \leq 23\lfloor \frac{t-1}{11} \rfloor$. Then there exists a menu selection problem with no (t, u)-stable menu. (cf. $u \geq 2t - 1 \Leftrightarrow$ existence when $g \leq 6$)

Set $x := \lfloor \frac{t-1}{11} \rfloor$. Then the following $70x$ agents have no (t, u)-stable menu:

#	Preference	#	Preference	#	Preference	#	Preference
$5x$	$1 \succ 2 \succ 3$	$3x$	$1 \succ 2 \succ 4 \succ 5$	x	$1 \succ 4 \succ 2 \succ 5$	x	$1 \succ 6 \succ 4 \succ 2$
$5x$	$2 \succ 3 \succ 4$	$3x$	$2 \succ 3 \succ 5 \succ 6$	x	$2 \succ 5 \succ 3 \succ 6$	x	$2 \succ 7 \succ 5 \succ 3$
$5x$	$3 \succ 4 \succ 5$	$3x$	$3 \succ 4 \succ 6 \succ 7$	x	$3 \succ 6 \succ 4 \succ 7$	x	$3 \succ 1 \succ 6 \succ 4$
$5x$	$4 \succ 5 \succ 6$	$3x$	$4 \succ 5 \succ 7 \succ 1$	x	$4 \succ 7 \succ 5 \succ 1$	x	$4 \succ 2 \succ 7 \succ 5$
$5x$	$5 \succ 6 \succ 7$	$3x$	$5 \succ 6 \succ 1 \succ 2$	x	$5 \succ 1 \succ 6 \succ 2$	x	$5 \succ 3 \succ 1 \succ 6$
$5x$	$6 \succ 7 \succ 1$	$3x$	$6 \succ 7 \succ 2 \succ 3$	x	$6 \succ 2 \succ 7 \succ 3$	x	$6 \succ 4 \succ 2 \succ 7$
$5x$	$7 \succ 1 \succ 2$	$3x$	$7 \succ 1 \succ 3 \succ 4$	x	$7 \succ 3 \succ 1 \succ 4$	x	$7 \succ 5 \succ 3 \succ 1$

- Simplified and cleaned from counterexample found by SMT solver.
- Also have somewhat improved upper bound: $u \geq (g-2)(t-1) + 1$.
- When $g \geq 7$, existence question open for $\underbrace{23\left\lfloor \frac{t-1}{11} \right\rfloor < u \leq (g-2)(t-1)}_{\sim g \text{ gap}}$.

Strategyproofness

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Fix g, t, u such that every menu selection problem has a stable menu.

Does there exist a strategyproof mechanism

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Theorem. For $g = 3, 4, 5, 6$, there is no anonymous SP mechanism.

Proof sketch. Given voting problem, carefully transform into menu selection problem and invoke Gibbard–Statterthwaite (transform so that unanimity implied by stability). Challenge: menu selection problem should only have singletons as stable menus.

Takeaways

- We introduce a new model for a matching market, with complementarities and no capacity constraints.
- For $g \leq 6$, we provide a tight characterization for when stable menus exist.
- For $g \geq 7$, we provide lower and upper bounds for when stable menus exist.
- For $3 \leq g \leq 6$, there are fundamental barriers for strategyproofness.

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Thank you! Questions?